DATA ANALYTICS PROJECT

ReLife – Returned Product Second-Life Analytics

"Every Return Deserves a Second Chance."

1. Project Title

- ReLife – Returned Product Second-Life Analytics

2. Product Name

- ReLife
- Tagline: "Every Return Deserves a Second Chance."

Main Problem Statement

Businesses need an effective way to analyze returned products and determine the most valuable next life for each product instead of treating every return as waste.

4. Proposed Solution

ReLife analyzes returned-product data and provides a data-driven recommendation for the product's next stage. The system analyzes product condition, return reason, original price, repair cost, estimated resale value, demand level, and storage cost.

Possible actions: Resell → Repair → Donate → Reuse Parts → Recycle

Unique Feature – Second-Life Score: Each returned product receives a score based on its condition, repair cost, resale value, demand, and storage cost. The score helps identify the best next action and the potential value that can be recovered.

5. Target / End Users

- Retail businesses
- E-commerce companies
- Electronics and appliance retailers
- Furniture and fashion businesses
- Warehouse and reverse-logistics teams
- Business managers and sustainability teams

6. Message to Users

"A returned product is not the end of its journey. Use data to find its best second life."

ReLife encourages businesses to recover value, reduce waste, and make smarter decisions about returned products.

7. Data Analysis

Analysis 1 – Returns by Product Category: Use a bar chart to compare the number of returned products across categories.

Analysis 2 – Return Reasons: Use a donut chart to show the proportion of products returned for different reasons such as damaged, defective, wrong product, size issue, or changed mind.

Analysis 3 – Return Trend: Use a line chart to show how the number of returns changes over time.

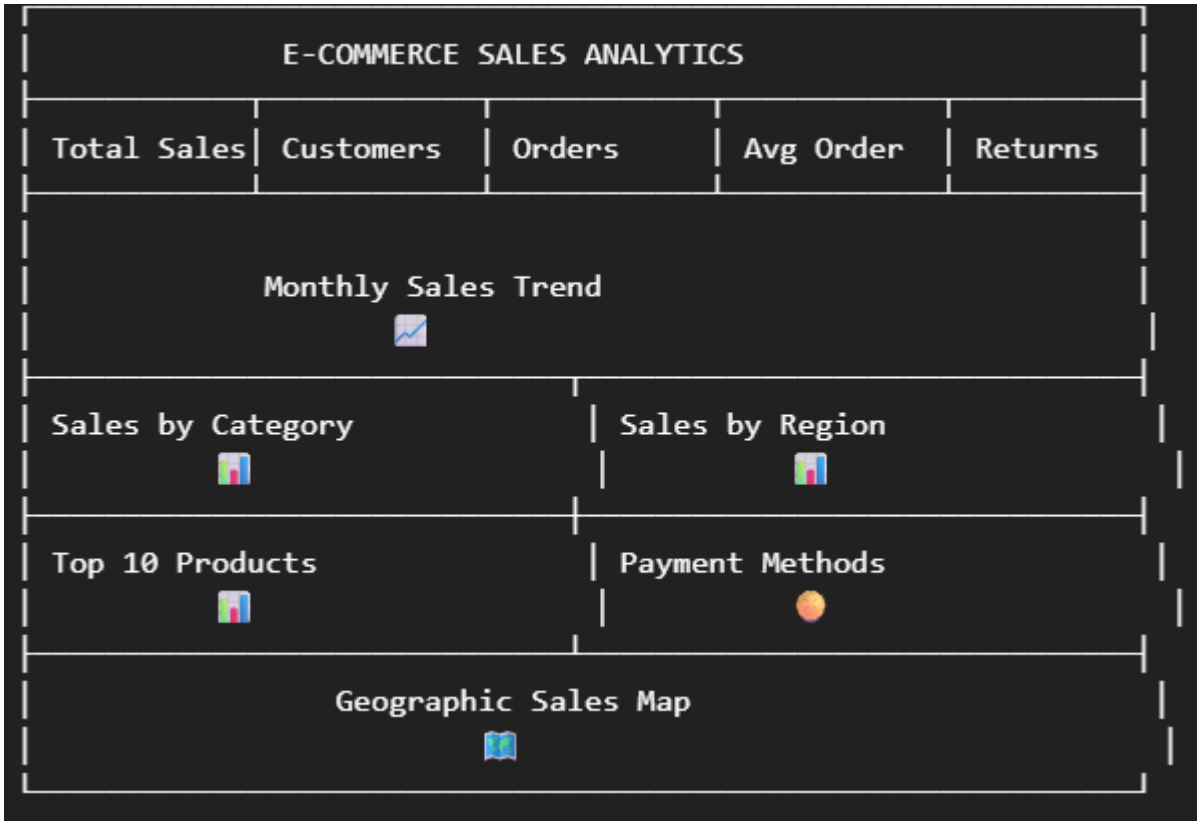
8. Second-Life Score

The Second-Life Score is the main analytical feature of ReLife. It considers product condition, estimated resale value, demand, repair cost, and storage cost.

Example:

- Product: Laptop A
- Condition: Good
- Repair Cost: ₹2,000
- Estimated Resale Value: ₹38,000
- Demand: High
- Second-Life Score: 88/100
- Recommendation: Repair & Resell

9. Dashboard Skeleton



Example search: Laptop A → display condition, repair cost, resale value, demand, Second-Life Score, and recommended action.

10. Dataset

- Kaggle: Product Returns Dataset Search
- Kaggle: E-commerce Returns Dataset Search
- Kaggle: Online Retail Dataset Search